

Table 1. Notation guide for the main quantities used in Section 3.

| Symbol                        | Meaning                                                                          |
|-------------------------------|----------------------------------------------------------------------------------|
| $T$                           | Number of tokens in the generated text.                                          |
| $T_s$                         | Number of sentences in the generated text.                                       |
| $p_t(\cdot)$                  | Baseline conditional distribution at step $t$ .                                  |
| $q_t(\cdot)$                  | Watermarked conditional distribution at step $t$ .                               |
| $P^s$                         | Sequence distribution induced by the baseline sampler.                           |
| $Q$                           | Sequence distribution induced by the watermarked sampler.                        |
| $\delta$                      | Bias strength in biased token-level sampling.                                    |
| $g_t$                         | Baseline green-set mass at step $t$ .                                            |
| $\hat{\sigma}^2$              | Variance term for bias-free token-level watermarking.                            |
| $Z_t$                         | Semantic evidence bit at sentence step $t$ .                                     |
| $\rho$                        | Semantic bias parameter controlling watermark strength.                          |
| $D_0$                         | Per-unit information budget before edits.                                        |
| $\text{TV}(P^s, Q)$           | Total variation between the baseline and watermarked distributions.              |
| $\text{KL}(Q \  P^s)$         | KL divergence from the watermarked distribution to the baseline distribution.    |
| $\varepsilon$                 | Token edit rate after post-generation editing.                                   |
| $\varepsilon_s$               | Probability that editing flips the semantic evidence bit.                        |
| $C_{\text{tok}}(\varepsilon)$ | Usable post-edit information budget for token-level watermarking.                |
| $C_{\text{sem}}(\varepsilon)$ | Usable post-edit information budget for semantic watermarking.                   |
| $\alpha$                      | False-alarm level of the detector.                                               |
| $\beta$                       | Target miss probability of the detector.                                         |
| $D_{\text{req}}^{\text{tok}}$ | Required information budget for token-level schemes under the given edit regime. |
| $D_{\text{req}}^{\text{sem}}$ | Required information budget for semantic schemes under the given edit regime.    |
| $\ell$                        | Average number of tokens per sentence.                                           |